Surface reflectance time series for CHESS

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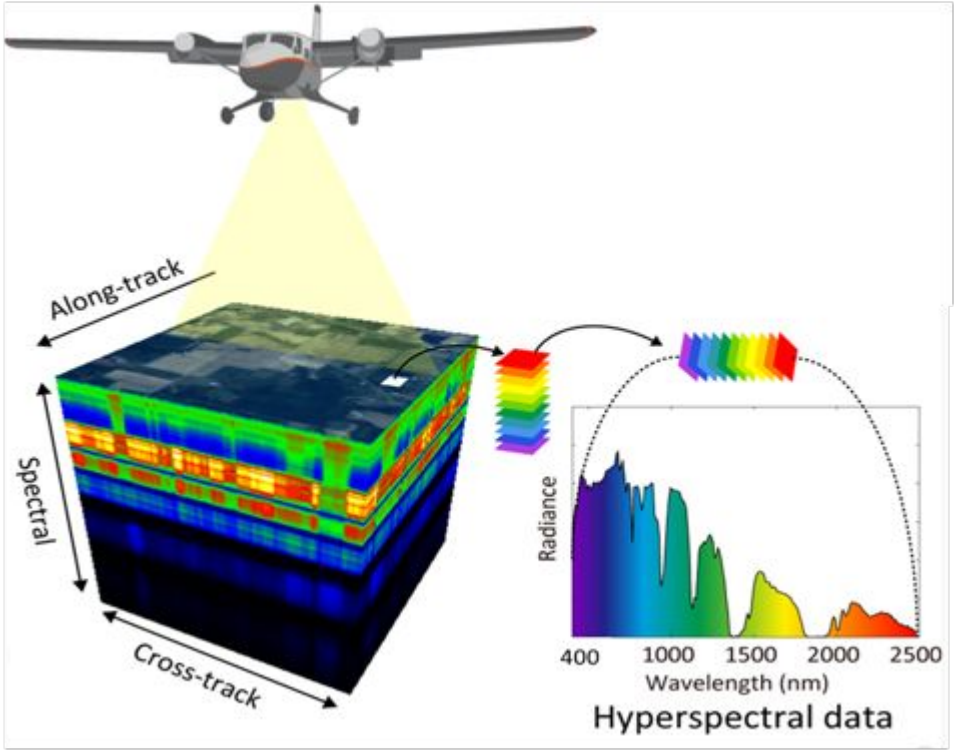
NEON Imaging Spectrometer (NIS)

380-2510 nm

426 bands

~5 nm spectral resolution

~1 m spatial resolution



https://www.neonscience.org/data-collection/imaging-spectrometer

Overview

NEON

JPL

2018 — 2025

raw DNs

raw DNs

v1.0

v2.0

At-sensor radiance
doi:10.15485/1617204

At-sensor radiance
doi:10.15485/3017966

At-sensor radiance
doi:10.15485/3017965

v1.0

v2.0

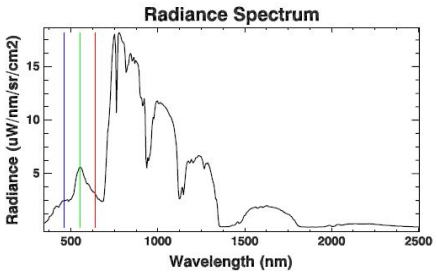
Surface reflectance
doi:10.15485/1618131

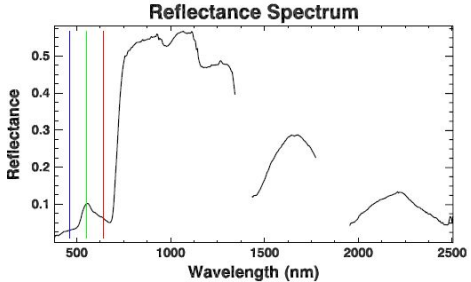
Surface reflectance
doi:10.15485/3013527

Surface reflectance
doi:10.15485/3013535

ACORN

ISOFIT





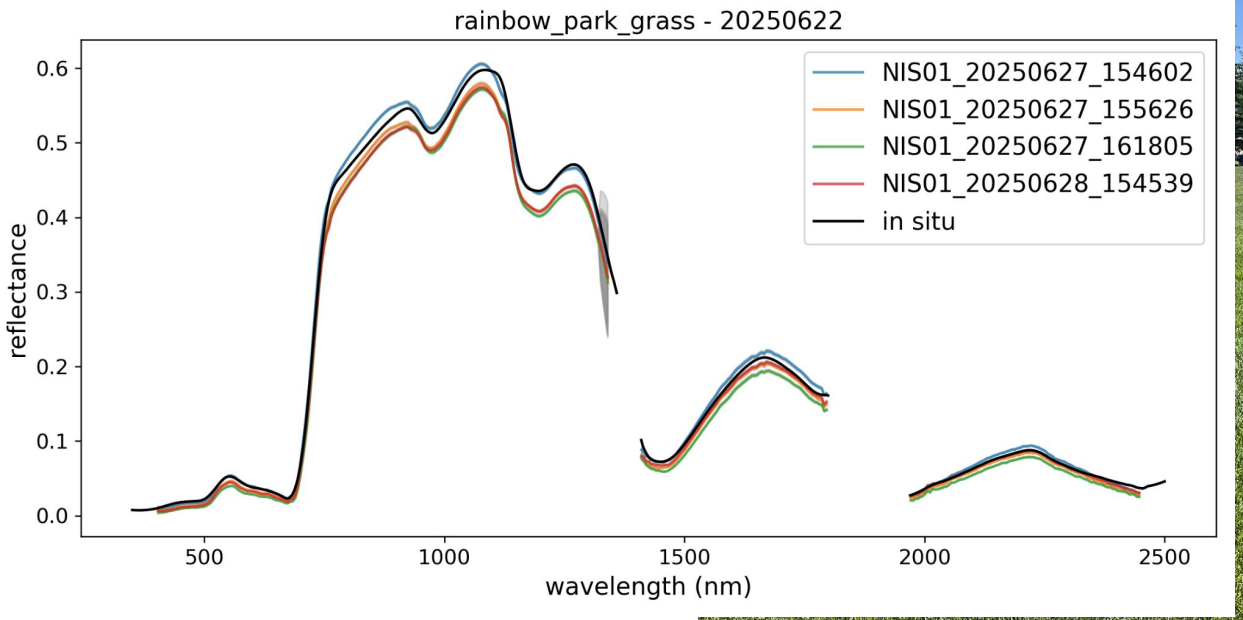
Why create a custom reflectance product?

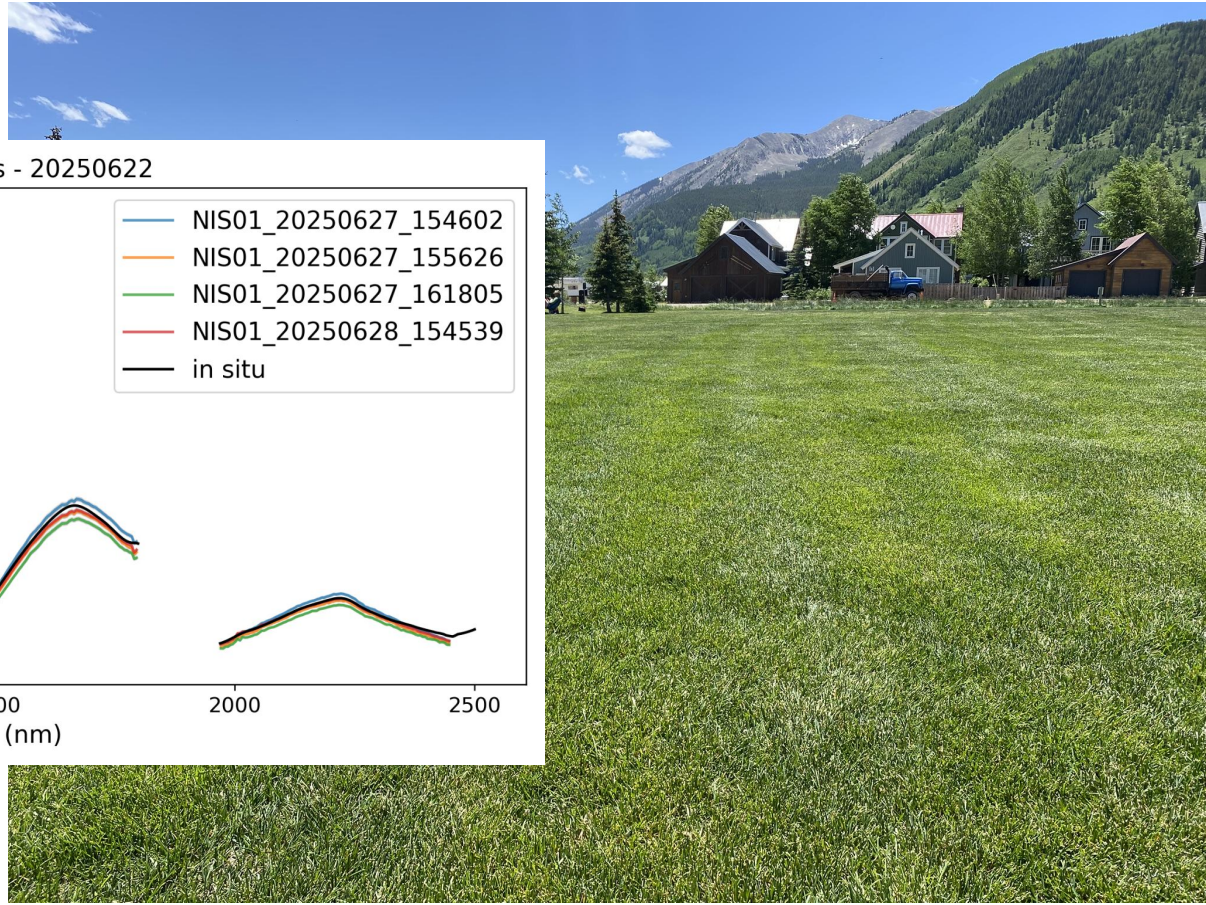
- Identical processing across years
- Recent advances in atmospheric correction from ISOFIT
  - Uncertainty!
- Vicarious radiance calibration

Process overview

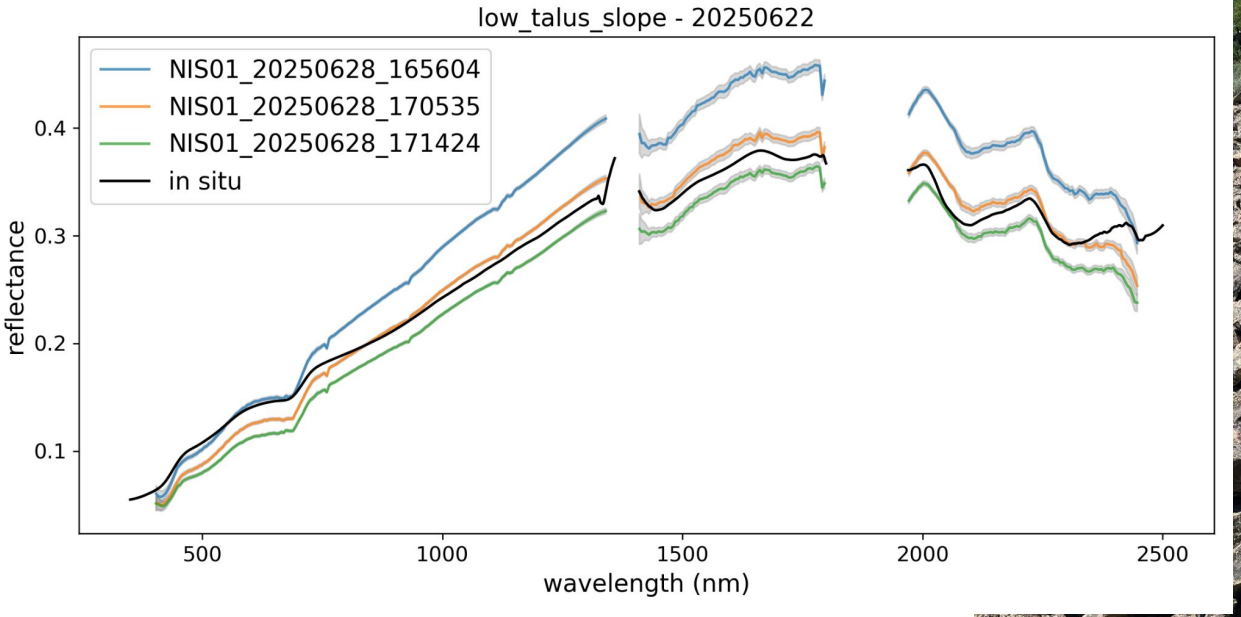
https://github.com/erin-carroll/chess-isofit/

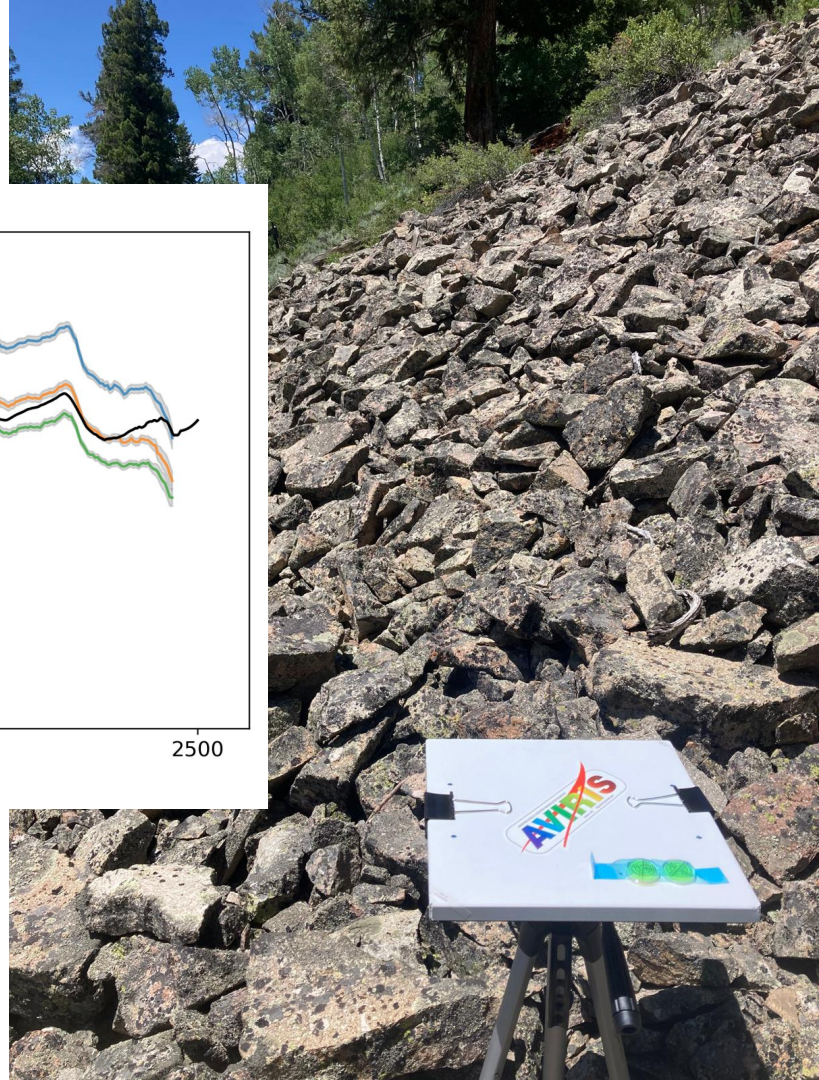
- Derive vicarious radiometric calibration coefficients (rccs)
- Run the isofit atmospheric correction per flightline
  - Develop instrument models
  - Six-component radiative transfer emulator (sRTMnet)
    - Topographic correction
- Calculate EWT
- Calculate shade
- Mosaic


rainbow_park_grass - 20250622
NIS01_20250627_154602
NIS01_20250627_155626
NIS01_20250627_161805
NIS01_20250628_154539
in situ
reflectance
0.0
0.1
0.2
0.3
0.4
0.5
0.6
500
1000
1500
2000
2500
wavelength (nm)



MAE = 0.011


low_talus_slope - 20250622
NIS01_20250628_165604
NIS01_20250628_170535
NIS01_20250628_171424
in situ
reflectance
0.4
0.3
0.2
0.1
500
1000
1500
2000
2500
wavelength (nm)


AVIRIS

MAE = 0.026

Custom surface reflectance, shade mask, and equivalent water thickness mosaics for the Colorado Headwaters Ecological Spectroscopy Study (2025)

```
ALMO/
    ├─ reflectance/
        ├─ flightline/
            ├─ NIS01_20250613_111111.zip
                ├─ NIS01_20250613_111111_rfl
                ├─ NIS01_20250613_111111_rfl.hdr
                ├─ NIS01_20250613_111111_uncert
                ├─ NIS01_20250613_111111_uncert.hdr
            ├─ …
        ├─ mosaic/
            ├─ ALMO_2025_mosaic_rfl_337000_4277000.nc
            ├─ ALMO_2025_mosaic_rfl_337000_4278000.nc
            ├─ ALMO_2025_mosaic_rfl_337000_4279000.nc
            ├─ …
    ├─ spatial/
        ├─ mosaic_boundary.geojson
        ├─ flightline_boundaries.geojson
    ├─ ALMO_2025_mosaic_ewt.tif
    ├─ ALMO_2025_mosaic_obs.tif
    ├─ ALMO_2025_mosaic_rfl_rgb.tif
    ├─ ALMO_2025_mosaic_shade.tif
CRBU/ …
UPTA/ …
```

```
├─ reflectance/
    ├─ flightline/
        ├─ NIS01_20250613_160117.zip
            ├─ NIS01_20250613_160117_rfl
            ├─ NIS01_20250613_160117_rfl.hdr
            ├─ NIS01_20250613_160117_uncert
            ├─ NIS01_20250613_160117_uncert.hdr
        ├─ …
    ├─ mosaic/
        ├─ ALMO_2025_mosaic_rfl_337000_4277000.nc
        ├─ ALMO_2025_mosaic_rfl_337000_4278000.nc
        ├─ ALMO_2025_mosaic_rfl_337000_4279000.nc
        ├─ …
```

Tutorials

Open in Colab

The mosaicked reflectance data are stored as netcdf files in 1000x1000 m chunks, on the same spatial grid as the NEON lidar-derived data products. Each netcdf f
426 bands measured by the NEON Imaging Spectrometer. The wavelength centers and full width at half maximum (fwhm) for each band in nanometers (nm) are

The file naming convention is:

`{domain}_{year}_rfl_{x coordinate of the lower left corner}_{y coordinate of the lower left corner}.nc`

This tutorial goes through opening a single .nc tile, exploring the data inside, and visualizing the data interactively alongside ancillary spatial data.

```
import xarray as xr
import geopandas as gpd
import numpy as np
import matplotlib.pyplot as plt
```

Thank you!

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EMIT Extended Mission Phase E Science

Berkeley
UNIVERSITY OF CALIFORNIA

RMBL
science. OUTSIDE.